

Erratum to MasterCube

Zheng Gong¹, Di Li¹,
Yufeng Tang², Dachao Wang³, Liangju Zhao^{1†}

¹ School of Computer Science, South China Normal University, Guangzhou, China
cis.gong@gmail.com, edu.lidi97@gmail.com, liangjzhao@126.com

² Institute for Network Sciences and Cyberspace, Tsinghua University, Beijing, China
yufengtang@tsinghua.edu.cn

³ Department of Electrical and Information Technology, Lund University, Lund, Sweden
mr.ongor@gmail.com

1 Original Article Metadata

- **Original Title:** MasterCube: A Family of Cross-Platform Hash Functions with AndRX Operations
- **Original Document Version:** Document Version 2.0 (Released: June 2026)
- **Corrected Document Version:** Document Version 2.1 (Released: September 2026)

2 Summary of Modifications

The modifications between Version 2.0 and Version 2.1 are classified into three categories. The first two categories represent the primary errata concerning the algorithm specification and the reference implementation, while the third category reports the updated implementation performance results.

2.1 Corrections to the Specification

Table 1 summarizes the corrections made to the algorithm specification and theoretical rationale in Sections 2 and 3 of the document.

2.2 Corrections to the Implementation

Table 2 summarizes the corrections made to the reference implementation and the resulting updates to the test vectors in Appendix A.

3 Implementation performance update

3.1 Updated Performance Results

Table 3 summarizes the updated ASIC synthesis results of the hardware architectures on the 45 nm Nangate cell library in Section 5.4.

[†]The authors are listed in alphabetic order.

Table 1: Corrections to the Specification.

Location	Original / Incorrect (v2.0)	Corrected / Should read (v2.1)
Page 7 , Section 2.1.3, Para. 1	$RC[i] = [0x243F \dots 0x7344]$ (big-endian order printed)	$RC[i] = [0x7344 \dots 0x243F]$, with clarified little-endian byte ordering.
Page 8 , Section 2.1.4, Para. 2	Omitted formal notation definition for inverse of MAndRX.	Added: “In Algorithm 2, $MAndRX^{-1}(A)$ denotes the inverse of $MAndRX(A)$.”
Page 8 , Algorithm 2, Lines 5–6	Swap slices, then $MAndRX(A)$ without inverse symbol.	Replaced with $A \leftarrow MAndRX^{-1}(A)$, and removed redundant slice swap.
Page 9 , Section 2.2, Para. 2 & List	“Only two operations need special attention, which are MAndRX and ShiftRows.”	“Only three operations need special attention, which are MAndRX, ShiftRows, and MixColumns.” Added note for MixColumns.
Page 9 , Section 2.2, Para. 3	(Omitted explanation of round constants distinction in ZIP).	Added explicit statement on using distinct round constants for Cube-p and Cube-p-ir .
Page 9 , Algorithm 4, Line 7	$A' \leftarrow \text{Cube-p-ir}(A', RC[i])$	$A' \leftarrow \text{Cube-p-ir}(A', RC[17 - i])$
Page 10 , Section 3.1, Para. 2	(Omitted edge case behavior of multi-rate padding).	Clarified that Keccak multi-rate padding can produce one more block in boundary cases.
Page 11–12 , Section 3.3, Paras. 1–3	Typo: “slices A_{xy}^0 and A_{xy}^0 ”, and omitted MixColumns inverse relation.	Fixed typo to “slices A_{xy}^0 and A_{xy}^1 ”, and explained that $M_{MixColumns}^1$ and $M_{MixColumns}^2$ are inverses.

Table 2: Corrections to the Implementation and Test Vectors.

Target	Original Issue (v2.0)	Correction in v2.1
Padding Rule	Multi-rate padding did not strictly append the terminating bit in certain block alignments, and length calculation had an integer overflow risk.	Corrected to strictly follow NIST/Keccak 10*1 padding, and resolved the buffer length overflow bug.
Inverse Round Swapping	Omitted the second slice swap in the software implementation of the inverse round function.	Added the required second slice swap to ensure exact algebraic inversion matching the specification.
Page 25 , Appendix A, Test vectors	Output hash digests in test vectors reflected the buggy padding and slice swap implementation.	Updated all test vectors for MasterCube-512, MasterCube-768, and MasterCube-1024.

Table 3: Updated Hardware Implementation Synthesis Results (MasterCube-512).

Architecture	Original Report (v2.0)	Updated Result (v2.1)	Impact
Round-Sequential (Table 14, Page 23)	Area = 94,713 μm^2 GE = 118,689 Delay = 1.54 ns Power = 10,122 μW	Area = 94,867 μm^2 GE = 118,880 Delay = 1.62 ns Power = 10,142 μW	Slight increase in area (+0.16%) and delay (+0.08 ns). Cycle counts unchanged.
Parallel Phase-Sliced (Table 15, Page 24)	Area = 102,798 μm^2 GE = 128,821 Delay = 1.66 ns Power = 55,559 μW	Area = 103,559 μm^2 GE = 129,773 Delay = 1.94 ns Power = 16,935 μW	Area (+0.74%) and delay (+0.28 ns), with substantial dynamic power reduction. Cycle counts unchanged.

3.2 Updates to Hardware Implementation Synthesis

In Section 5.4, the synthesis results of the hardware architectures on the 45 nm Nangate cell library were updated following revisions to the RTL designs.

1. Round-Sequential Implementation (Table 14). For the round-sequential architecture of MasterCube-512, the synthesis results are updated as follows:

- **Original (v2.0):** Area = 94,713 μm^2 , Gate Equivalents (GE) = 118,689, Critical Path Delay = 1.54 ns, Dynamic Power = 10,122 μW .
- **Corrected (v2.1):** Area = 94,867 μm^2 , Gate Equivalents (GE) = 118,880, Critical Path Delay = 1.62 ns, Dynamic Power = 10,142 μW .

2. Parallel Phase-Sliced Implementation (Table 15). For the parallel phase-sliced architecture of MasterCube-512, the synthesis results are updated as follows:

- **Original (v2.0):** Area = 102,798 μm^2 , Gate Equivalents (GE) = 128,821, Critical Path Delay = 1.66 ns, Dynamic Power = 55,559 μW .
- **Corrected (v2.1):** Area = 103,559 μm^2 , Gate Equivalents (GE) = 129,773, Critical Path Delay = 1.94 ns, Dynamic Power = 16,935 μW .

The cycle counts for both implementations remain identical to the previous report.

Specification errata. Table 4 records corrections to the written specification and pseudocode.

Implementation errata. Table 5 records code corrections and the resulting test-vector update.

Implementation performance update. We measured v2.0 and v2.1 at eight message lengths from 32 to 65536 bytes. At each point, the relative change is $100(C_{2.1}/C_{2.0} - 1)\%$, where C is CPU cycles per byte. For the optimized software implementations, each table entry summarizes 24 new/old cycle-per-byte ratios; negative changes mean fewer cycles in v2.1. Across all 120 optimized-software points the largest absolute change is 4.31%. Thus the measured software performance differences are small for these platforms and lengths. The ARM report records different fixed-input Reference digest fingerprints

Table 4: Specification errata in MasterCube v2.1. These entries correct the written description and pseudocode; implementation corrections are listed separately.

ID	Location	v2.0 issue	v2.1 correction
S1	Round constants	The displayed constant order did not match the intended representation.	State the little-endian order used for the constants.
S2	Algorithm 2, inverse round	The pseudocode wrote MAndRX instead of its inverse and included a redundant swap.	Write MAndRX^{-1} , explain the notation, and remove that swap.
S3	Section 2.2, inverse operations	The explanation named only MAndRX and ShiftRows as needing special handling.	Include MixColumns and explain the slice-swap mechanism.
S4	Section 2.2, round constants	The forward and inverse branch constants were not clearly distinguished.	Clarify their respective use in the two branches.
S5	Algorithm 4, inverse branch	The inverse-round constant index was written incorrectly.	Use $RC[17 - i]$ in the inverse branch.

Table 5: Implementation errata in MasterCube v2.1. The revised test vectors are a consequence of the implementation corrections.

ID	Component	v2.0 issue	v2.1 correction and effect
I1	Message padding	The code did not implement the specified Keccak multi-rate padding in all cases.	Correct padding to follow the specified pad10*1 rule, including the edge case requiring an extra block.
I2	Message length	Integer overflow could occur while calculating the message length.	Correct the length calculation and its boundary handling.
I3	Inverse round	The slice-swap sequence omitted a second swap.	Add the missing swap so the inverse-round state is arranged as specified.
I4	Test vectors	Outputs based on the old implementation no longer describe v2.1.	Regenerate the vectors in the code package and document; this is a consequence of I1–I3, not a separate algorithm change.

Table 6: Implementation Performance Update: v2.1 relative to v2.0. Software entries summarize CPU cycles per byte over 24 points per implementation (three digest lengths and eight message lengths); ASIC entries compare synthesis metrics with the revised ASIC report’s v2.0 baseline. Negative software values indicate fewer cycles.

<i>Software: CPU cycles per byte</i>				
Platform	Implementation	Geometric mean	Minimum	Maximum
x86-64	Performance-optimized	+0.63%	-0.51%	+2.21%
x86-64	Resource-optimized	-0.73%	-2.97%	+0.72%
x86-64	AVX512	+0.43%	-0.21%	+1.14%
ARM	Performance-optimized	-0.78%	-2.68%	+1.94%
ARM	Resource-optimized	-3.71%	-4.31%	-0.14%
<i>ASIC: synthesis metric change</i>				
Design	Area	GE	Delay	Power
Round-sequential	+0.16%	+0.16%	+5.19%	+0.20%
Parallel phase-sliced	+1.10%	+1.10%	+1.57%	+1.51%

at all 24 checked settings; this is a comparison of corrected algorithm versions, not a pure optimization of identical outputs. Reference and SHA3-512 are excluded from the performance version comparison. The ASIC percentages use the v2.0 baseline in the revised synthesis report, not the previously printed parallel-design row.

References

- [1] Keccak Team. The Keccak Reference and Specification. <https://keccak.team/keccak.html>.
- [2] A. Flórez-Gutiérrez, L. Grassi, G. Leander, F. Sibleyras, and Y. Todo. General practical cryptanalysis of the sum of round-reduced block ciphers and ZIP-AES. In *Advances in Cryptology – ASIACRYPT 2024*, LNCS 15486, pages 345–375. Springer, 2024.